SeizureFormer: A Multi-Scale Transformer for Seizure Risk Forecasting from RNS-Derived Biomarkers

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AFFILIATIONS & FUNDS

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01. Introduction

Epilepsy is a chronic neurological disorder characterized by spontaneous and recurrent seizures, driven by complex interactions between cortical excitability and multi-timescale rhythms. Continuous intracranial recordings from Responsive Neurostimulation (RNS) systems provide stable, interpretable biomarkers of brain activity—namely interictal epileptiform activity (IEA) and long episodes (LEs)—that reflect underlying network excitability. However, long-horizon seizure forecasting using these structured signals remains largely unexplored.

02. Objective

To address this gap, we propose SeizureFormer, a multi-scale Transformer model that integrates CNN patch embedding, cross-variable temporal convolution, and squeeze-and-excitation attention to jointly capture short-term fluctuations and long-term seizure cycles. By leveraging RNS-derived biomarkers rather than raw EEG, SeizureFormer enables clinically actionable, 1–14-day seizure-risk prediction, supporting proactive and personalized management in epilepsy care.

Related literature

- Most prior work uses scalp EEG for seizure forecasting — limited by noise and short-term focus.
- RNS systems provide stable intracranial biomarkers (IEA, Long Episodes) linked to cortical excitability.
- Studies show circadian rhythms modulate seizure risk (Baud et al., Lancet Neurol., 2018).
- Transformer-based temporal models excel in multivariate biosignal forecasting.

03. Methodology

- Inputs: Daily RNS biomarkers (IEA, LE)
- CNN + temporal convolution for local and cross-channel patterns
- Transformer + SE attention for long-range cycles
- Output: 1–14 day seizure-risk forecast

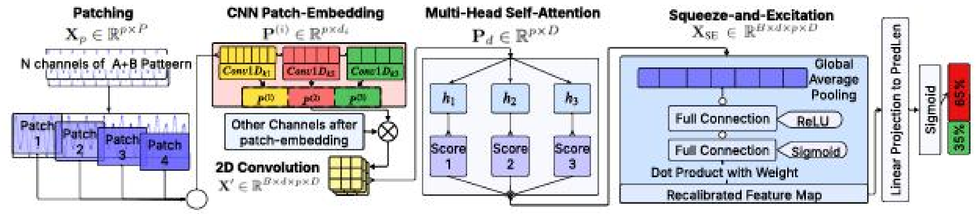

Figure 1: Overall framework of SeizureFormer.

04. Results/Findings

- State-of-the-art performance: Mean ROC AUC = 79.44%, PR AUC = 76.29% across five patients and four horizons.
- Outperforms ML & DL baselines
- Stable across patients

Table 2: Ablation experiment results on Patient 0477. The best performance is bold and the second best is underlined.

Model Variant	PredLen = 1		PredLen = 3		PredLen = 7	
Metrics	ROC AUC	PR AUC	ROC AUC	PR AUC	ROC AUC	PR AUC
Full Model	**84.83%**	**73.03%**	**80.62%**	**80.70%**	81.63%	**90.70%**
w/o CNN Patch Embedding	84.39%	71.89%	73.31%	75.60%	64.61%	81.59%
w/o SE Block	82.80%	68.68%	77.95%	78.21%	**82.76%**	90.20%
w/o Cross-Variable Temporal convolution	77.71%	63.55%	77.66%	77.24%	75.69%	88.65%
w/o All Modules	76.35%	61.86%	70.87%	72.40%	72.97%	87.56%

05. Analysis

- IEA and LE rhythms reveal clear circadian and infradian periodicity aligned with seizure risk.
- Feature attention maps highlight sensitivity to biomarker surges preceding high-risk days.
- Ablation results confirm complementary contributions of CNN, convolution, and Transformer layers.
- Demonstrates potential for multi-day clinical planning and adaptive neurostimulation scheduling.

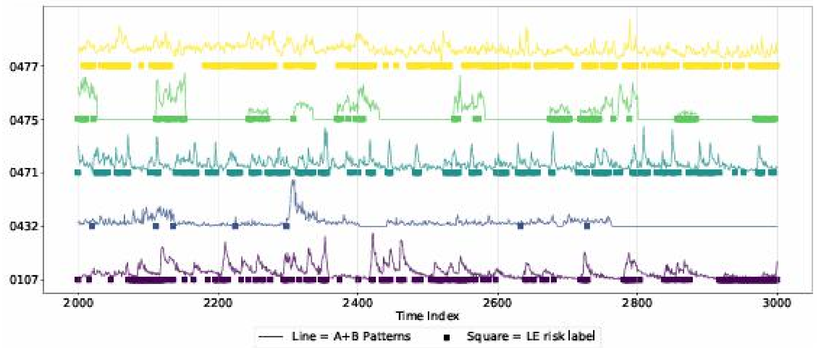

Figure 2: A+B patterns and LE risk labels over a 1000-day period. The line represents normalized A+B patterns, while the square markers indicate high risk labels.

06. Conclusion

- SeizureFormer enables reliable 1–14-day seizure-risk forecasting from stable RNS biomarkers.
- Integrates multi-scale temporal modeling for both short-term and long-term dynamics.
- Demonstrates strong generalizability and clinical relevance for personalized epilepsy management.
- Paves the way toward biomarker-driven, adaptive neurostimulation and proactive seizure prevention.

Charts

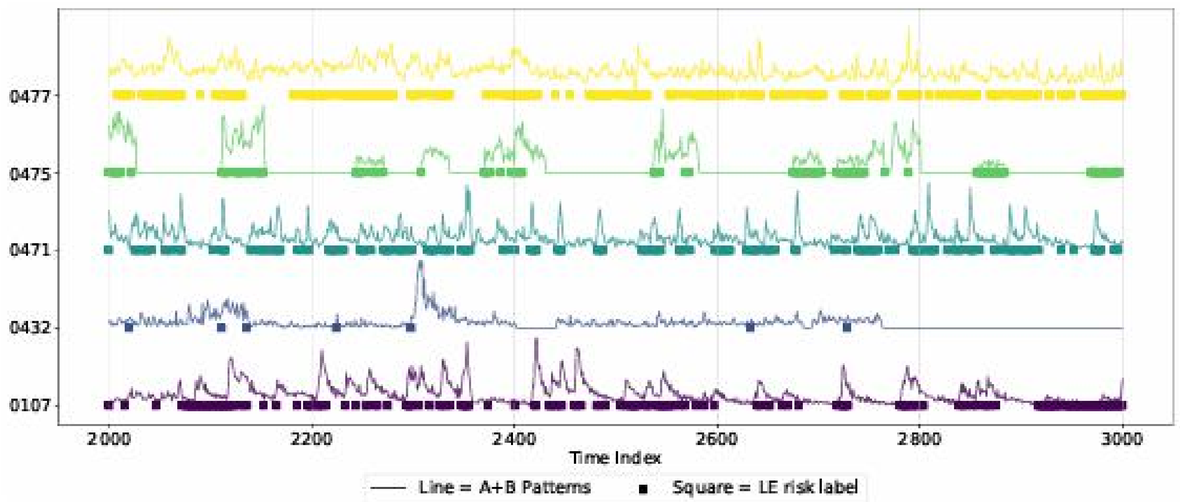

Figure 2: A+B patterns and LE risk labels over a 1000-day period. The line represents normalized A+B patterns, while the square markers indicate high risk labels.

Patients		0107		0432		0471		0475		0477		Average	
Models	Pred Lengths	ROC AUC	PR AUC	ROC AUC	PR AUC	ROC AUC	PR AUC	ROC AUC	PR AUC	ROC AUC	PR AUC	ROC AUC	PR AUC
GLM	1	59.5%	37.6%	77.8%	19.6%	68.9%	55.9%	95.9%	88.7%	81.2%	67.1%	71.32%	64.34%
	3	55.2%	51.5%	69.8%	21.8%	67.6%	76.3%	92.5%	86.8%	76.8%	75.8%		
	7	58.7%	70.2%	53.4%	25.9%	67.5%	87.3%	90.2%	87.0%	80.4%	88.4%		
	14	58.2%	81.7%	21.5%	69.4%	95.3%	86.4%	84.1%	NA	NA	NA		
Poisson Regression	1	62.7%	48.8%	78.0%	27.1%	67.7%	54.7%	96.2%	90.8%	80.8%	65.9%	71.20%	65.24%
	3	54.6%	51.3%	78.8%	33.5%	63.0%	73.6%	93.6%	88.5%	71.5%	69.4%		
	7	59.8%	70.6%	72.1%	31.2%	59.6%	85.7%	90.5%	87.6%	76.9%	87.2%		
	14	65.0%	82.8%	38.2%	17.2%	61.6%	93.7%	82.2%	79.9%	NA	NA		
Logistic Regression	1	62.4%	44.9%	82.3%	22.3%	68.4%	54.8%	94.8%	87.1%	79.3%	62.6%	73.13%	64.94%
	3	53.9%	49.1%	77.9%	28.2%	65.3%	73.8%	93.2%	87.6%	76.6%	74.6%		
	7	55.3%	66.7%	70.9%	31.7%	68.7%	88.1%	90.4%	87.4%	81.3%	89.2%		
	14	59.1%	82.2%	54.3%	25.3%	69.8%	95.3%	85.6%	82.9%	NA	NA		
SVM	1	60.4%	39.4%	64.4%	16.4%	68.8%	55.2%	95.8%	88.5%	79.3%	63.0%	71.35%	63.92%
	3	55.5%	51.3%	72.9%	22.9%	68.1%	76.1%	93.3%	87.9%	75.5%	71.8%		
	7	57.3%	68.4%	62.1%	27.0%	67.5%	88.0%	90.6%	87.4%	81.3%	89.2%		
	14	57.7%	81.4%	50.8%	22.9%	69.0%	95.2%	85.4%	82.4%	NA	NA		
DLinear	1	70.9%	57.7%	46.2%	8.96%	68.9%	55.9%	96.9%	92.0%	81.8%	67.6%	72.83%	65.97%
	3	52.9%	52.9%	73.4%	22.7%	69.4%	78.0%	93.6%	88.9%	79.3%	78.6%		
	7	47.1%	61.6%	84.2%	37.2%	69.1%	89.5%	88.2%	85.1%	79.0%	89.6%		
	14	52.5%	77.5%	80.6%	35.6%	69.5%	95.2%	80.3%	78.9%	NA	NA		
GRU_LSTM	1	58.2%	81.7%	82.5%	32.4%	68.5%	56.3%	**97.6%**	93.3%	83.9%	68.4%	73.16%	71.05%
	3	67.9%	66.2%	87.8%	54.3%	70.8%	78.7%	94.7%	88.9%	76.7%	73.5%		
	7	64.5%	75.1%	76.8%	30.8%	58.4%	88.1%	90.2%	86.2%	50.0%	85.2%		
	14	45.6%	77.6%	35.1%	51.9%	51.9%	85.7%	82.6%	NA	NA	NA		
PatchTST	1	**74.2%**	63.9%	93.2%	62.4%	66.4%	46.0%	97.1%	91.7%	78.2%	62.7%	76.41%	73.07%
	3	52.0%	45.4%	92.6%	59.6%	69.6%	76.4%	94.5%	89.9%	77.4%	78.7%		
	7	53.7%	70.8%	87.4%	49.5%	73.1%	91.0%	91.2%	88.0%	78.8%	89.2%		
	14	28.9%	64.3%	83.9%	**78.5%**	**74.6%**	**95.9%**	85.0%	84.5%	NA	NA		
SeizureFormer	1	**74.2%**	53.5%	**94.7%**	61.6%	70.3%	59.2%	97.1%	**93.4%**	**84.8%**	73.0%	**79.44%**	**76.29%**
	3	61.4%	68.1%	92.2%	77.8%	71.0%	76.7%	94.8%	90.8%	80.6%	80.7%		
	7	47.7%	64.6%	87.9%	52.3%	71.8%	90.6%	92.7%	88.3%	81.6%	**90.7%**		
	14	64.7%	**83.1%**	85.5%	64.6%	73.3%	95.7%	86.0%	84.9%	NA	NA		

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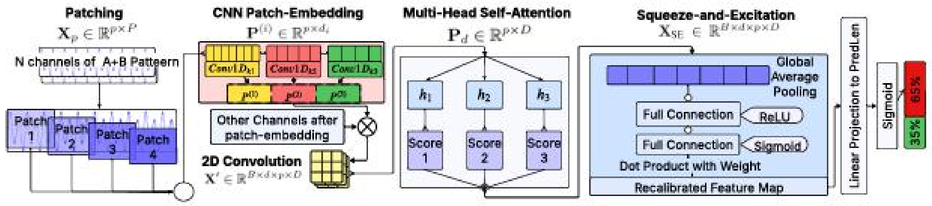

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